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$$f(x) = \ln\left(\sqrt[3]{3x + \sqrt{9x^2+1}}\right)$$
$$f'(x) = (\ln(u))' \cdot (u)'$$
$$= \frac{1}{u} \cdot \left(\sqrt[3]{3x + \sqrt{9x^2+1}}\right)'$$
$$= \frac{1}{u} \cdot \frac{1}{3\left(3x + \sqrt{9x^2+1}\right)^{\frac{2}{3}}} \cdot \left(3 + \frac{18x}{2\sqrt{9x^2+1}}\right)$$
$$= \frac{1}{u} \cdot \frac{3 + \frac{9x}{\sqrt{9x^2+1}}}{3\left(3x + \sqrt{9x^2+1}\right)^{\frac{2}{3}}}$$
$$= \frac{1}{u} \cdot \frac{\frac{3\sqrt{9x^2+1} + 9x}{\sqrt{9x^2+1}}}{3\left(3x + \sqrt{9x^2+1}\right)^{\frac{2}{3}}}$$
$$= \frac{1}{u} \cdot \frac{3\sqrt{9x^2+1} + 9x}{3\sqrt{9x^2+1}\left(3x + \sqrt{9x^2+1}\right)^{\frac{2}{3}}}$$
$$= \frac{1}{u} \cdot \frac{\sqrt[3]{3x + \sqrt{9x^2+1}}}{\sqrt{9x^2+1}}$$
$$= \frac{1}{\sqrt[3]{3x + \sqrt{9x^2+1}}} \cdot \frac{\sqrt[3]{3x + \sqrt{9x^2+1}}}{\sqrt{9x^2+1}}$$
$$= \frac{1}{\sqrt{9x^2+1}}$$

References